Debrato Ghosh

debrato2005@gmail.com · github.com/Debrato2005 · linkedin.com/in/debrato-ghosh

Education

Manipal Institute of Technology Manipal, India
Bachelor of Technology in Information Technology, CGPA: 8.66/10 Aug 2024 – May 2028
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Computer Networks, Operating Systems, Object-Oriented Programming
Sardar Vallabhbhai Patel Vidyalaya and Jr. College Mumbai, India
Class XII (HSC) – 86.83% 2023
Kapol Vidyanidhi International School Mumbai, India
Class X (ICSE) – 96.83% 2021

Technical Skills

Languages: C, C++, Java, Python, JavaScript, SQL
Backend & APIs: FastAPI, Flask, REST APIs, Node.js, SQLAlchemy, Pydantic, JWT/OAuth
Frontend: React, HTML5, CSS3, JavaFX
Databases & Caching: PostgreSQL, MySQL, Redis
DevOps & Systems: Docker, Git, Linux/Unix, Nginx
Machine Learning: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch
Computational Biology: pVACtools/pVACseq, MHCflurry, PyClone-VI, Cancer Genomics
Tools: VS Code, Eclipse, GitHub, Postman, pgAdmin

Experience

Research Intern – 2D/3D Computer Vision | aiRender Technology Pvt. Ltd. Jul 2026 – Present
• Researching deep learning-based 2D-to-3D human reconstruction for a real-time 3D video conferencing platform
• Investigating model, inference and latency optimizations for real-time streaming of reconstructed 3D video
Machine Learning & Backend Development Intern | Madtales.ai Jul 2025 – Aug 2025
• Built CNN–GRU and BERT-based document-understanding pipelines, achieving 85%+ line classification accuracy and reducing structured data extraction errors by 30%
• Designed a Flask REST API with TensorFlow Lite integration, cutting mobile inference latency by 40%; architected an async TTS system with cloud storage processing 10k+ requests

Projects

OncoSyn – Computational Oncology & Immunotherapy R&D Platform | Founder Aug 2026 – Present
• Building an evolution-aware, heterogeneity-aware, and spatially informed computational oncology platform for multimodal tumour-state modelling and precision immunotherapy R&D.
• Architecting neoantigen portfolio optimization, TCR-T/CAR-T co-design, spatial/single-cell evidence integration, uncertainty modelling, and adaptive experiment prioritization.
• Designing pipelines around pVACtools, MHCflurry, PyClone-VI, TRACERx/TESLA datasets, and CP-SAT optimization with reproducible provenance, sensitivity benchmarking, and evidence-gated research decisions.
Smolink – Production-Grade URL Shortening Platform (In Progress) Jun 2026 – Present | GitHub
• Building a production-oriented URL shortener with FastAPI, async SQLAlchemy, PostgreSQL and Redis using layered architecture, strict TDD and horizontally scalable design
• Implemented Snowflake/Base62 short codes, custom aliases, authenticated ownership and atomic Redis sliding-window rate limiting with guest/user quotas
• Engineered secure authentication with Argon2id, email verification, JWT access/refresh tokens, refresh-token rotation and

replay detection, password reset and Google OAuth/OIDC integration

UniSpace – Web-Based Classroom Booking & Scheduling System Apr 2026 | GitHub

• Architected a normalized PostgreSQL schema with triggers, stored procedures and optimistic concurrency control

• Engineered a three-tier Java application with JDBC, Servlets and JSP featuring role-based authentication and session management

• Optimized PostgreSQL queries achieving sub-100ms response times while supporting 50+ concurrent reservation requests

Additional Projects: Cryptalis – Python/SQLAlchemy application-layer encrypted data-protection platform; OrbitOps – real-time 3D satellite tracking and collision-risk analysis; StaySync – JavaFX MVC hotel-management system with persistence and multithreading

Leadership & Extracurricular Activities

Computational Astrophotography – Processed 300+ astronomical frames across 15+ deep-sky objects using calibration, registration and stacking workflows (Siril, DSS, Sequator)

PAV Head – PRISM, Manipal Institute of Technology

Student Clubs – Photography Club & The Astronomy Club, Manipal; organized and hosted image-processing workshops

Member–Volunteer Services Organization

Public edition · Phone number omitted · Source: owner-supplied resume